

AdventureWorks

Advanced Sales Forecasting & Simulation — Executive Report

Probabilistic 12-month sales forecast for annual planning

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1. Executive Summary

AdventureWorks asked for a risk-aware alternative to single-point sales forecasts: not just what next year's sales will be, but the full range of plausible outcomes and the probability of hitting financial targets. Two simulation models were built on the company's complete transaction history (July 2017 – June 2020, 121,253 order lines, \$109.8M of sales), plus extensions that correct the weaknesses of the second model and exploit the star-schema dimensions for segment-level forecasts.

- **Model 1:** a top-down Monte Carlo simulation of the annual growth rate (10,000 trials), and
- **Model 2:** a bottom-up bootstrap that resamples historical daily sales (1,000 simulated years). A month-stratified variant restores seasonality and current sales levels; a joint-day variant adds channel, category and territory forecasts that are exactly consistent with the total. (An optional transaction-level simulation, adapted from the class demonstration, is used only as a diagnostic cross-check.)

Headline results (next 12 months, June 2020 – June 2021):

- If the historical growth trend continues, expected sales are \$75.9M, with a 90% confidence range of \$68.2M – \$83.6M (Monte Carlo).
- Under a conservative zero-growth assumption, sales stay at the current run-rate: \$53.0M, range \$50.5M – \$55.5M (stratified bootstrap on the last 365 days).
- Exceeding last year's sales by 10% is close to certain if the trend holds, but has only a ~12% probability if growth stalls completely — the gap between these two numbers is the company's real planning risk.
- Segment view (zero-growth scenario): Internet \$16.0M / Reseller \$37.0M; Bikes \$44.4M (84% of revenue); Europe is the growth engine — 25% of recent revenue versus an 18% three-year average share.

Recommendation: use \$68M – \$76M as the main planning range (budget floor to expected value under trend), with \$53M as the downside stress case and \$84M as the upside capacity case. Details in Section 8.

2. Business Problem and Data

The leadership team needs a probabilistic sales forecast for the upcoming 12-month period to support strategic resource allocation and risk management. This report follows the IMPACT cycle (identify the question, master the data, perform the test plan, address and refine results, communicate insights, track outcomes).

2.1 Data preparation and validation

The source workbook (AdventureWorks Sales.xlsx) is a dimensional star schema: Sales_data is the fact table and Date, Product, Customer, Reseller and Territory are dimensions. Validation findings before any analysis:

- The fact table contains 121,253 order lines totalling \$109,809,274 in sales.
- The date dimension is a pre-built calendar running to 30 June 2021, but the last actual transaction is on 15 June 2020 — the usable history is therefore 1 July 2017 to 15 June 2020.
- All 1,081 days in that window contain at least one transaction: no gap-filling was needed.
- The abrupt end on 15 June 2020 is a data-extraction boundary, not a business collapse: the final day records 447 order lines (~\$373k, one of the strongest days on file), and the last 20 days average \$210,541/day versus \$151,247/day over the prior 90 days. The series is truncated while activity is still rising.
- Only 2018 and 2019 are complete calendar years; 2017 covers July–December and 2020 covers January to mid-June. This has a direct consequence for the growth-rate methodology (Section 3.2).

How the numbers are computed — Daily sales series: convert OrderDateKey (a yyyymmdd integer) to a date, group the 121,253 order lines by order date, and sum the Sales Amount column. This yields 1,081 daily totals covering 2017-07-01 to 2020-06-15 with no missing days. The truncation check compares the mean of the last 20 daily totals with the mean of the 90 daily totals before them. Every model in this report is built from this one series.

3. Historical Sales Profile

3.1 Trend and seasonality

Two features of the history drive every modelling decision: a strong upward trend — sales roughly quadruple across the three years — and clear seasonality within each year. A simulation that ignores the trend will under-forecast; one that ignores seasonality cannot produce a realistic daily profile.

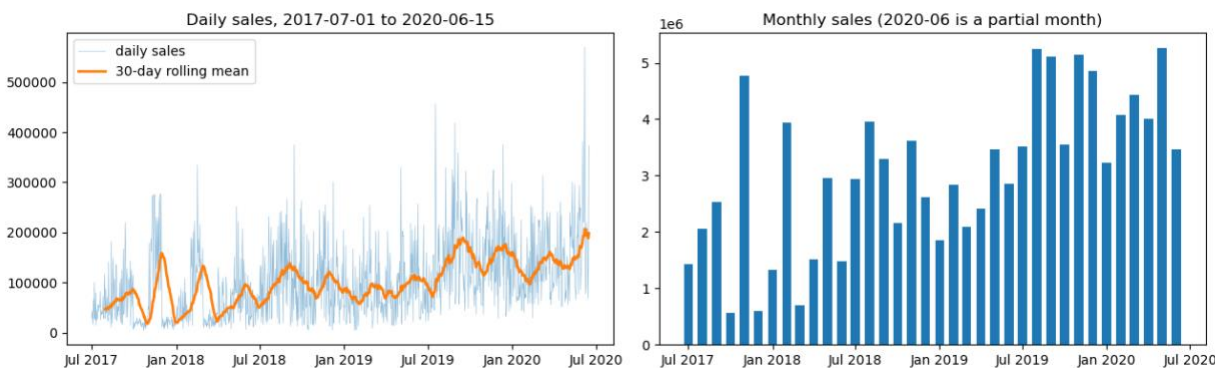


Figure 1 — Daily sales with 30-day rolling mean (left) and monthly totals (right). The last monthly bar is the partial month June 2020.

Table 1 — Sales by calendar year.

Calendar year	Total sales	Completeness
2017	\$11.93M	partial — July to December only
2018	\$30.52M	complete
2019	\$42.90M	complete
2020	\$24.47M	partial — 1 January to 15 June

How the numbers are computed — Calendar-year totals: sum the daily series within each calendar year. The 2020 figure covers only 166 days, which is why it cannot be compared directly with 2018 or 2019.

3.2 The growth-rate problem and how it was solved

The Monte Carlo method requires a set of historical year-over-year growth rates from which to estimate a mean and standard deviation. Taking the brief literally — calendar-year growth rates — yields exactly one observation: 2019 vs 2018 = +40.6%. No standard deviation can be estimated from a single number. Three alternative definitions of annual growth were therefore evaluated:

Table 2 — Three candidate definitions of historical annual growth.

Option	Definition	n	Mean	Std dev	Assessment
A	Monthly YoY growth	23	80.6%	87.1%	Rejected for simulation: monthly-

					scale noise; low-base months in 2017–18 inflate the mean
B	Fiscal-year (Jul–Jun) YoY	2	47.5%	6.7%	Right quantity but only two observations; kept as a cross-check
C	Rolling fiscal-year (TTM) YoY	12	43.2%	8.8%	Chosen: annual-scale growth with enough observations for a meaningful spread

Option C is used as the primary growth distribution. Because the company's fiscal year runs July–June, option C is best read as a rolling fiscal year: every trailing-12-month window is a fiscal-year-length window, with the year-end rolled across the twelve month-ends — the fiscal-year logic of option B, extended to yield 12 observations instead of 2. Option B itself is doubly fragile: with $n = 2$ the standard deviation is just the gap between two numbers, and it swings from 6.7% to 11.8% depending solely on how the 15 missing June days of FY2020 are imputed. Option C's known limitation is stated openly: rolling windows overlap, so the 12 observations are positively correlated and the standard deviation is likely somewhat understated — one reason the sensitivity analysis in Section 4.3 stress-tests the volatility assumption.

How the numbers are computed — Option A: each month's total divided by the same month one year earlier, minus 1 (23 observations after dropping the partial June 2020). Option B: fiscal-year totals (July–June) compared year over year (2 observations). Option C: $TTM(t) = \text{sum of the 12 months ending at month } t$; $\text{growth} = TTM(t) / TTM(t-12) - 1$, evaluated each month from June 2019 to May 2020 (12 observations). Means are arithmetic averages; standard deviations are sample standard deviations ($n-1$ denominator).

4. Part 1 — Aggregate Monte Carlo Simulation (Top-Down)

4.1 Methodology and assumptions

1. Estimate the distribution of annual growth from the 12 rolling fiscal-year (TTM) observations: mean $\mu = 43.2\%$, standard deviation $\sigma = 8.8\%$.
2. Heuristic assumption (stated explicitly): future annual growth follows a Normal distribution $N(\mu = 0.432, \sigma = 0.088)$.
3. Forecast base: the most recent complete 365-day window (17 June 2019 – 15 June 2020), totalling \$53,024,764. The most recent complete calendar year (2019) is six months stale and would understate the current run-rate; the sensitivity analysis quantifies exactly what the literal reading would cost.
4. Run 10,000 trials. Each trial draws one growth rate g from the Normal distribution and computes simulated sales = base $\times (1 + g)$.

4.2 Results

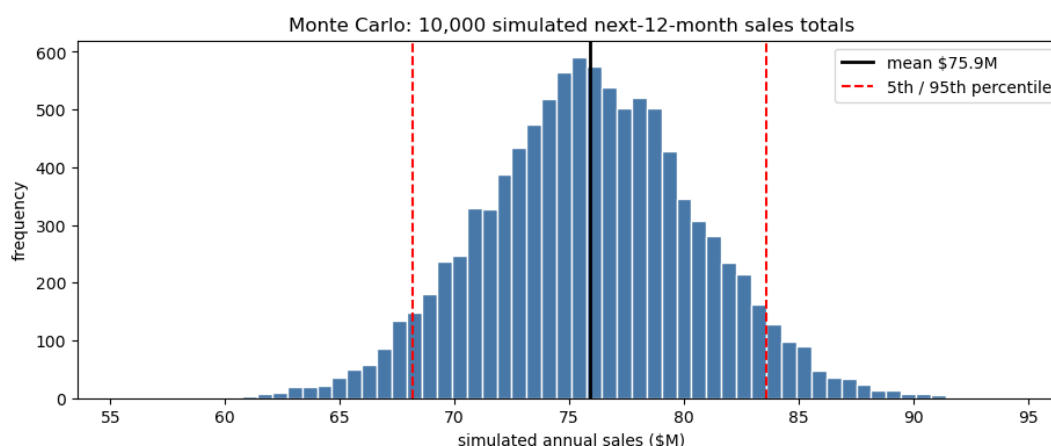


Figure 2 — Distribution of 10,000 simulated next-12-month sales totals.

- Expected (mean) sales: \$75.9M.
- 90% confidence interval: \$68.2M – \$83.6M (5th to 95th percentile).
- Probability of exceeding the base by 10% (\$58.3M): $\approx 100\%$. Under the fitted growth distribution this target sits almost five standard deviations below the mean — the brief's example target is simply not ambitious for this company. A more informative management target: the probability of exceeding \$80M is roughly 20%.

How the numbers are computed — Expected sales = arithmetic mean of the 10,000 trial outcomes. The 90% interval is the 5th and 95th percentiles of those outcomes. Target probabilities are the share of trials exceeding the threshold, e.g. $P(\text{sales} > \text{base} \times 1.10) = (\text{number of trials above } \$58.3\text{M}) / 10,000$. Random seed fixed at 42, so every figure is exactly reproducible.

4.3 Sensitivity of the results to the assumptions

Because the growth definition, the Normal heuristic and the forecast base are all modelling choices, the simulation was re-run under alternative assumptions:

Table 3 — Monte Carlo results under alternative assumptions.

Scenario	μ	σ	Mean	5th pct	95th pct	P(>base+10%)
C rolling FY (primary)	0.432	0.088	\$76.0M	\$68.4M	\$83.9M	≈100%
B fiscal-year (n = 2)	0.475	0.067	\$78.3M	\$72.3M	\$84.2M	≈100%
C with $\sigma \times 1.5$	0.432	0.132	\$75.9M	\$64.3M	\$87.4M	99.5%
Zero-growth stress	0.000	0.088	\$52.9M	\$45.3M	\$60.5M	12.1%
Literal brief: base = CY2019	0.432	0.088	\$61.4M	\$55.1M	\$67.6M	79.4%

The central forecast is robust to the choice between growth definitions B and C (means differ by ~3%). Two rows matter for decision-making. The zero-growth stress: if growth stops entirely, sales land near \$53M and the +10% target flips from near-certain to unlikely (12%). And the literal-brief row: applying growth to the stale calendar-2019 base (\$42.9M) instead of the current run-rate (\$53.0M) would cut the central forecast by \$14.6M — quantifying why the base was chosen deliberately.

How the numbers are computed — Rows 1–4 re-run the same 10,000-trial simulation with different (μ , σ) inputs; 'zero-growth stress' keeps historical volatility but sets expected growth to 0. Row 5 reuses the primary growth draws and rescales by base_2019 / base (a pure change of base), so it isolates the base effect. Its probability column keeps the same absolute target (\$58.3M) for comparability.

5. Part 2 — Daily Bootstrap Simulation (Bottom-Up)

5.1 Methodology

1. Build the historical pool: the 1,081 daily sales totals.
2. Simulate one future year: draw 365 values from the pool with replacement (a chosen day is returned to the pool and can be drawn again).
3. Repeat 1,000 times, giving a $1,000 \times 365$ matrix of simulated daily sales.
4. Summarise: per-day percentiles form the daily forecast band; row sums give 1,000 possible annual totals.

The method is fully non-parametric — no distributional assumption is made. Two built-in properties of this plain, fully random resampling must be understood when reading the results:

- Every future day is drawn from the same pool, so day 1 and day 200 are statistically identical: the daily band is flat and day-of-week or seasonal structure is not preserved (contrary to common intuition about bootstrapping).
- The pool mixes low-level 2017 days with high-level 2020 days, so simulated years centre on the average historical year — not on the company's current run-rate.

5.2 Results

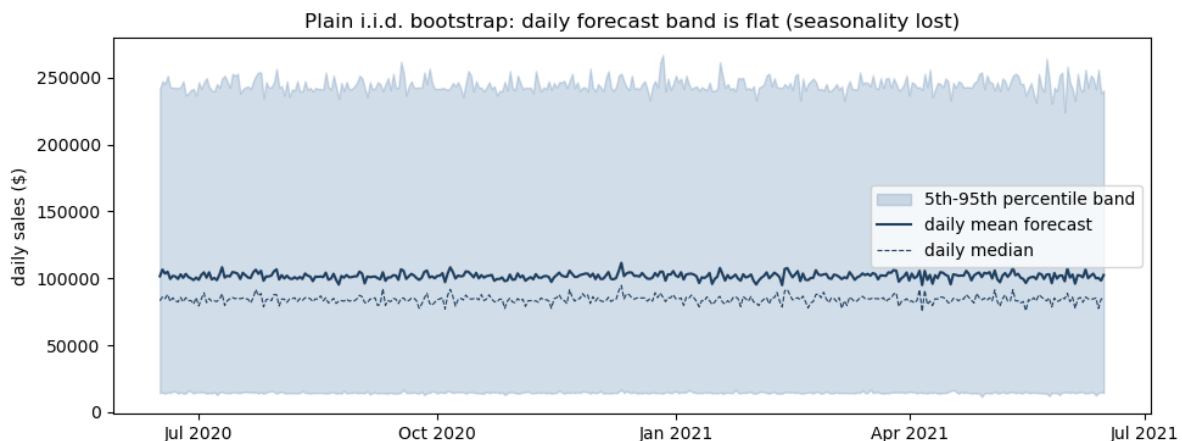


Figure 3 — Daily forecast band from the plain bootstrap. The band is flat: seasonality is not preserved by independent random resampling.

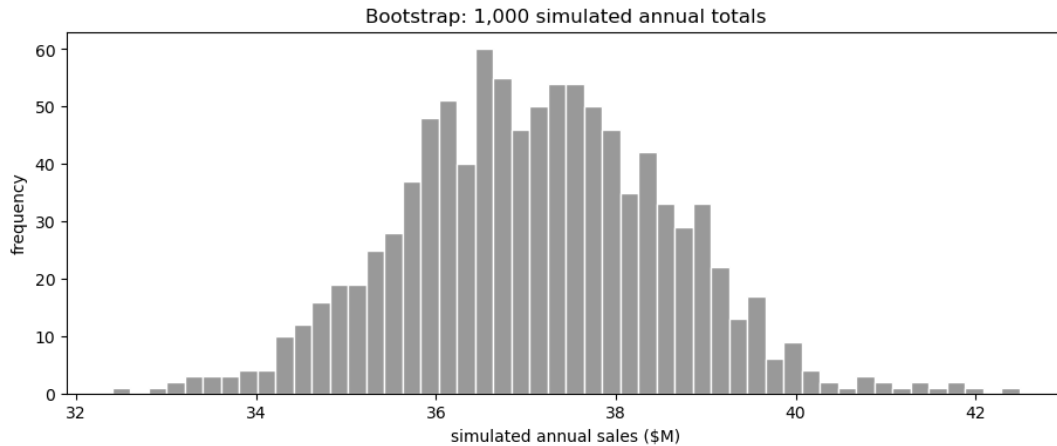


Figure 4 — Distribution of the 1,000 bootstrapped annual totals.

- Mean annual total: \$37.1M; 90% interval: \$34.7M – \$39.5M.
- The interval is strikingly narrow (width \$4.8M vs \$15.4M for the Monte Carlo). When 365 independently drawn days are added together, their ups and downs largely cancel out — a statistical averaging effect — so the range looks deceptively tight. The model treats good and bad days as independent events, ignoring that good and bad years are systematic: it understates true annual risk.

How the numbers are computed — Daily band: for each of the 365 forecast days, take the 5th, 50th and 95th percentiles of that day's 1,000 simulated values. Annual totals: sum each simulation's 365 daily values (row sums of the matrix), then take the mean and the 5th/95th percentiles of the 1,000 sums.

6. Extensions to the Bootstrap

6.1 Month-stratified bootstrap

Two refinements repair the weaknesses identified above while keeping the method fully non-parametric:

- **Stratify by calendar month:** A simulated day in month m is drawn only from historical days of the same calendar month, which restores the seasonal shape of the daily profile.
- **Restrict the pool to the last 365 days:** Drawing only from the most recent 365 days removes the downward bias caused by mixing 2017-level days with current-level days. The implicit assumption becomes 'next year looks like the last 12 months' (zero growth), making this variant directly comparable with the Monte Carlo base.

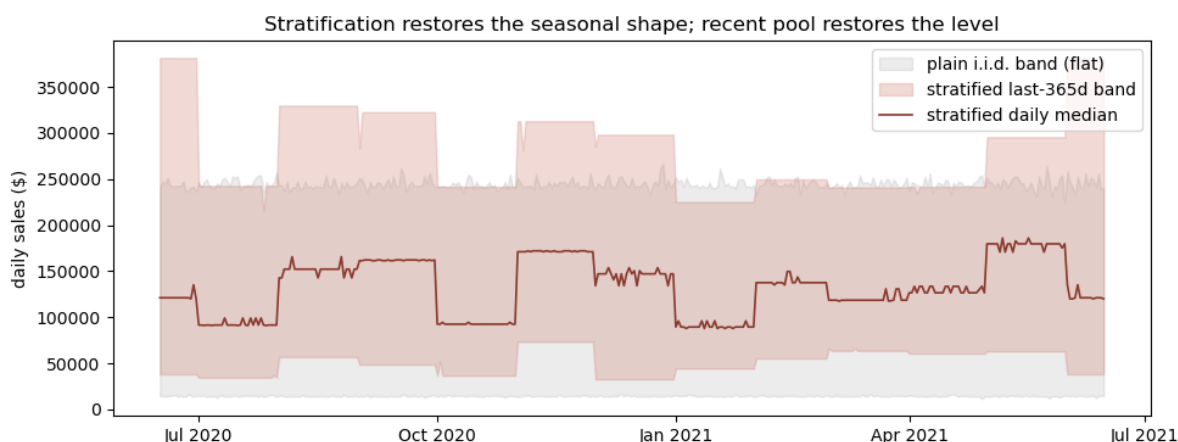


Figure 5 — The stratified last-365-day band (red) restores both the seasonal shape and the current sales level; the plain band (grey) is flat and too low.

- Stratified, all history: mean \$37.0M (90%: \$34.9M – \$39.5M) — level still biased down.
- Stratified, last 365 days: mean \$53.0M (90%: \$50.5M – \$55.5M) — matches the current run-rate and serves as the conservative zero-growth scenario.

Because the forecast days fall inside the company's pre-built fiscal calendar, the stratified forecast can be reported in fiscal quarters for planning:

Table 4 — Zero-growth forecast by fiscal quarter (stratified last-365-day bootstrap).

Fiscal quarter	Median	5th pct	95th pct
FY2020 Q4 (16–30 June only)	\$2.35M	\$1.63M	\$3.23M
FY2021 Q1	\$13.79M	\$12.60M	\$15.34M
FY2021 Q2	\$13.55M	\$12.40M	\$14.67M
FY2021 Q3	\$11.58M	\$10.67M	\$12.56M
FY2021 Q4 (to 15 June)	\$11.62M	\$10.57M	\$12.81M

How the numbers are computed — Stratification: twelve pools are built, one per calendar month; a forecast day dated July draws only from historical July days. The last-365-day variant builds its pools from 2019-06-17 to 2020-06-15 only (≈ 30 days per month). Quarterly figures are computed correctly: each simulation's daily values are first summed within the quarter, then the median and percentiles are taken across the 1,000 quarterly totals — summing daily percentiles would be wrong because percentiles are not additive.

6.2 Hierarchy-consistent joint-day bootstrap (channel, category, territory)

The star-schema dimensions invite segment-level forecasts. The naive route — one independent simulation per segment, then summing — is statistically unsafe here: segment dynamics are strongly cross-correlated (the two sales channels' rolling-year growth rates correlate at about -0.7 , Internet accelerating while Reseller decelerates), so independent segment simulations would mis-state the risk of the total.

Table 5 — Segment growth diagnostic (rolling fiscal-year growth over the full history; see notebook section 4c-i). Only a minority of segments are statistically estimable, and the channel growth rates correlate at -0.72 — the evidence for choosing joint-day resampling over per-segment simulations.

Dimension	Segment	Share	Growth mean	Growth σ	Usable for simulation?
Channel	Internet	27%	+66%	68%	No — early low base inflates mean and σ
Channel	Reseller	73%	+40%	12%	Yes, but covers only 70% of sales
Territory	Europe	18%	+187%	13%	No — low-base ramp-up artifact
Territory	North America	72%	+23%	6%	Yes
Territory	Pacific	10%	+77%	64%	No — small and volatile
Category	Bikes	86%	+37%	13%	Yes — but it is nearly the total itself
Category	Components	11%	+91%	78%	No — volatile, late ramp-up
Category	Accessories / Clothing	1–2%	+240–570%	—	No — pure low-base noise

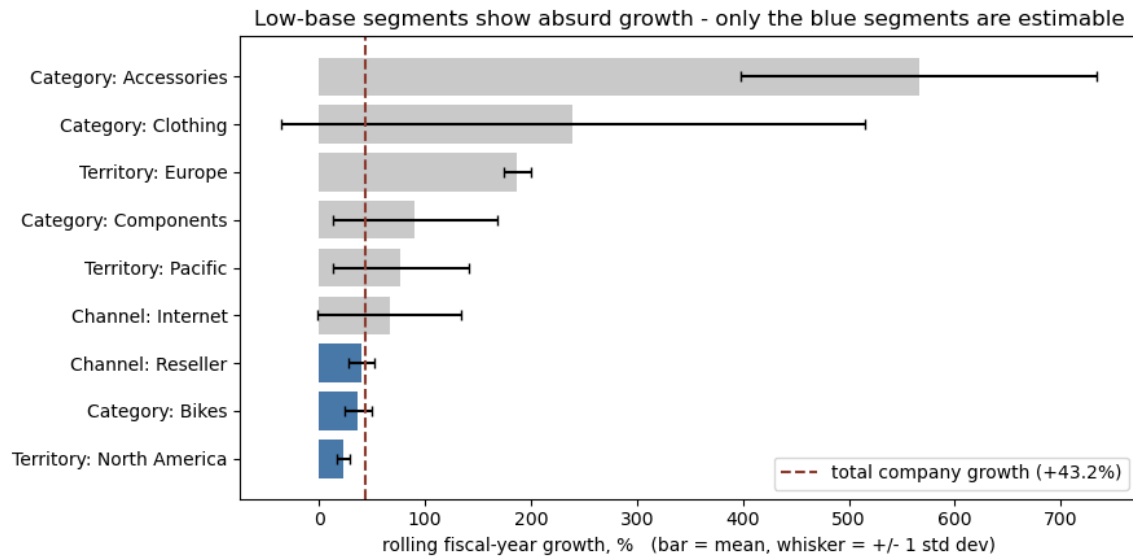


Figure 6 — Rolling fiscal-year growth by segment (bar = mean, whisker = ± 1 std dev). Low-base segments (grey) produce inflated, unstable growth rates; only the blue segments are statistically estimable. The dashed line marks total-company growth (+43.2%).

The fix costs one design change: resample day indices instead of day values. When a historical day is drawn, its entire segment decomposition travels with it, so all within-day cross-segment structure is preserved exactly — no correlation matrix to estimate — and every segment forecast is consistent by construction: channels, categories and territories add up to the simulated total in every single trial, so the numbers handed to channel and regional managers always reconcile with the company total.

To be precise about roles: the calendar month remains the only sampling stratum — it decides which historical days a future day may be drawn from. Channel, category and territory are not strata; they are decompositions carried along by each drawn day. Consequently, adding a reporting dimension changes the total's distribution not at all; it only adds rows to the output. The customer dimension is deliberately excluded: 18,486 customers with mostly one or two orders each would produce pure-noise forecast rows; the customer key is 'not applicable' for the entire reseller channel; and customer-level questions (retention, lifetime value) are a different analysis. Geographic customer structure is already captured by territory.

Table 6 — Coherent segment forecasts, zero-growth scenario (joint-day bootstrap).

Segment	Mean	5th pct	95th pct	Share
Total	\$53.04M	\$50.50M	\$55.54M	100%
Channel: Internet	\$16.00M	\$15.63M	\$16.35M	30%
Channel: Reseller	\$37.04M	\$34.49M	\$39.56M	70%
Category: Accessories	\$1.11M	\$1.07M	\$1.15M	2%
Category: Bikes	\$44.43M	\$42.39M	\$46.43M	84%
Category: Clothing	\$1.32M	\$1.25M	\$1.39M	2%
Category: Components	\$6.18M	\$5.69M	\$6.67M	12%

Territory: Europe	\$13.48M	\$12.28M	\$14.65M	25%
Territory: North America	\$33.73M	\$31.71M	\$35.79M	64%
Territory: Pacific	\$5.84M	\$5.46M	\$6.23M	11%

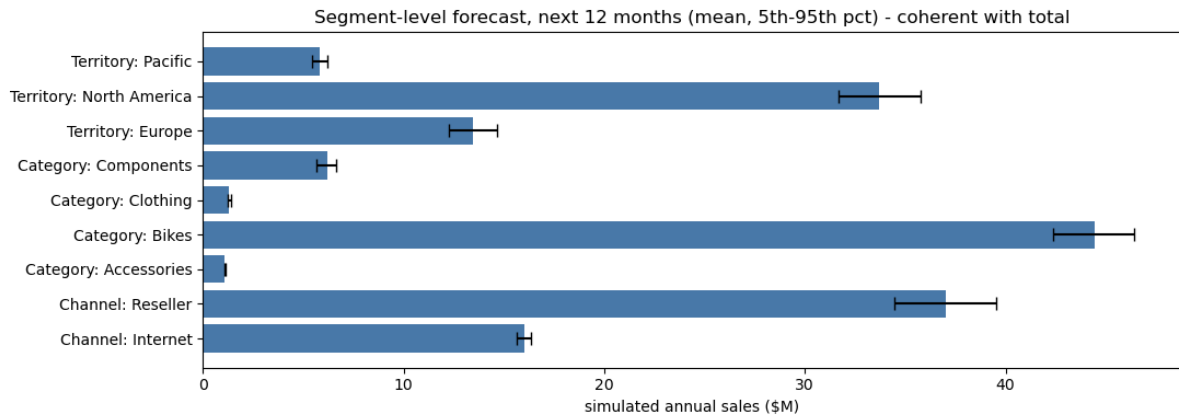


Figure 7 — Segment-level forecast (mean with 5th–95th percentile bars), coherent with the total by construction.

- Coherence verified in code: in all 1,000 trials, channel parts and territory parts each sum exactly to the simulated total.
- Business insight: Internet's interval is narrow (steady B2C demand) while Reseller's is wide (lumpy B2B batch orders) — the two channels need different planning buffers.
- Business insight: Europe contributes 25% of recent revenue versus an 18% three-year average share — it is the company's growth engine and merits its own target-setting.

How the numbers are computed — Same month-stratified sampling over the last-365-day pool as Section 6.1, but the draw is a day index. For each drawn day, the day's total and its decomposition by channel (Internet / Reseller, identified by ResellerKey = -1), product category (joined from Product_data) and territory group (joined from Sales Territory_data) are taken together. Segment annual totals are the sums of the segment's daily values over each simulated year; shares are segment means divided by the total mean. An independent random stream (seed 7) is used so this section does not perturb the other simulations.

6.3 Optional extension — transaction-level Poisson simulation (diagnostic cross-check)

This section is a diagnostic supplement to the two required models, not a third headline forecast: its role is to validate the zero-growth results at the finest grain and to illustrate how resampling granularity affects risk estimates.

The instructor's demonstration notebook resamples at the finest grain: individual order lines. Its original form draws a daily transaction count from a single Poisson rate ($\lambda \approx 112$ lines/day, the average over the full three-year history) and then samples that many order lines from the entire fact table. Two of the weaknesses already identified apply: one global λ has no seasonality, and pooling all of history drags the level down to the three-year average. The adaptation used here fixes both:

- **Monthly rates:** the average number of order lines per day is measured separately for each calendar month of the most recent year: rates range from ~194 lines/day (July) to ~273 (May), versus a single all-history rate of 112 — which understates current activity by half.
- **Recent, month-matched pool:** Order lines are drawn only from the same calendar month of the last 365 days, carrying their product category tags.
- **Computational shortcut:** a convenient property of the Poisson distribution lets one monthly draw replace thirty daily draws — the results are mathematically identical to the instructor's day-by-day version, but ~41 million simulated rows cheaper (details in the grey note below).

Table 7 — Category forecasts from the transaction-level simulation (1,000 simulated years).

Category	Mean	5th pct	95th pct	Share
Accessories	\$1.11M	\$1.09M	\$1.13M	2.1%
Bikes	\$44.50M	\$43.79M	\$45.20M	83.7%
Clothing	\$1.32M	\$1.29M	\$1.36M	2.5%
Components	\$6.20M	\$6.03M	\$6.38M	11.7%

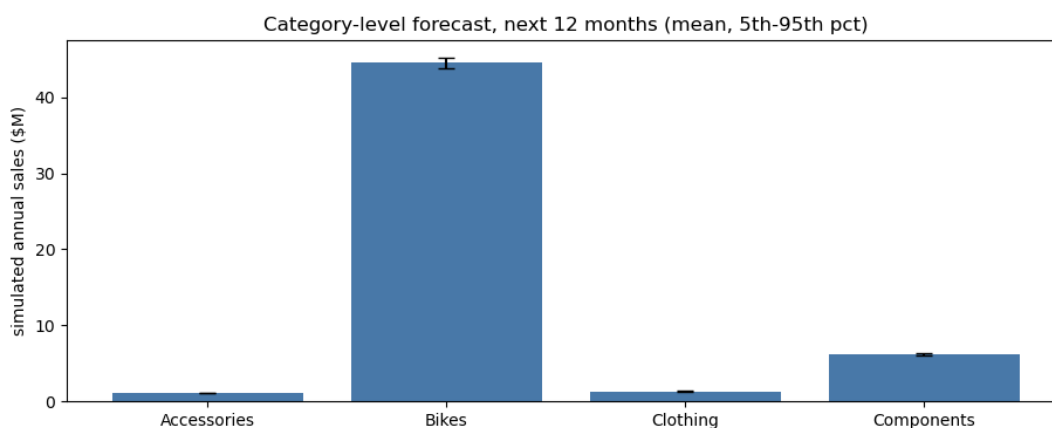


Figure 8 — Category-level forecast from the transaction-level simulation.

- Three-way cross-validation: the transaction-level annual total (\$53.14M) agrees with the stratified daily bootstrap (\$53.00M) and the joint-day bootstrap (\$53.04M) to within 0.3% — three independently coded resampling schemes over the same zero-growth pool converge on the same answer.
- Honest caveat: sampling order lines independently breaks up multi-line B2B orders that in reality arrive together on one day. This erases within-day clustering, so the 90% interval shrinks to \$52.4M – \$53.9M (width \$1.5M) — even narrower than the daily bootstrap's already-narrow \$5.0M. The finer the resampling unit, the more a bottom-up simulation understates annual-level risk.

How the numbers are computed — For each simulation and month m : draw the month's transaction count $N \sim \text{Poisson}(\lambda_m \times \text{number of forecast days in } m)$, where $\lambda_m = (\text{order lines of month } m \text{ in the last-365-day pool}) \div (\text{days of month } m \text{ in that pool})$; then draw N order lines with replacement from that month's pool and add their Sales Amount into the simulation's category totals. Category means/percentiles are computed across the 1,000 simulated years. The sample mean is the standard (maximum-likelihood) estimate of a Poisson rate, and the sum

of D independent $\text{Poisson}(\lambda)$ day-counts is $\text{Poisson}(D \times \lambda)$ — the identity that lets one monthly draw replace thirty daily draws.

7. Model Comparison

Table 8 — All simulated distributions of the next-12-month sales total. The joint-day bootstrap matches 'stratified, last 365 days' by design and is omitted. The transaction-level row is an extension shown for diagnosis only, not a headline model.

Model	Mean	5th pct	95th pct	Width	P(>base+10%)
Monte Carlo (trend-following)	\$75.9M	\$68.2M	\$83.6M	\$15.4M	≈100%
Plain bootstrap (as specified)	\$37.1M	\$34.7M	\$39.5M	\$4.8M	0%
Stratified, all history	\$37.0M	\$34.9M	\$39.5M	\$4.6M	0%
Stratified, last 365 days	\$53.0M	\$50.5M	\$55.5M	\$5.0M	0%
Transaction-level (extension — diagnostic only)	\$53.1M	\$52.4M	\$53.9M	\$1.5M	0%

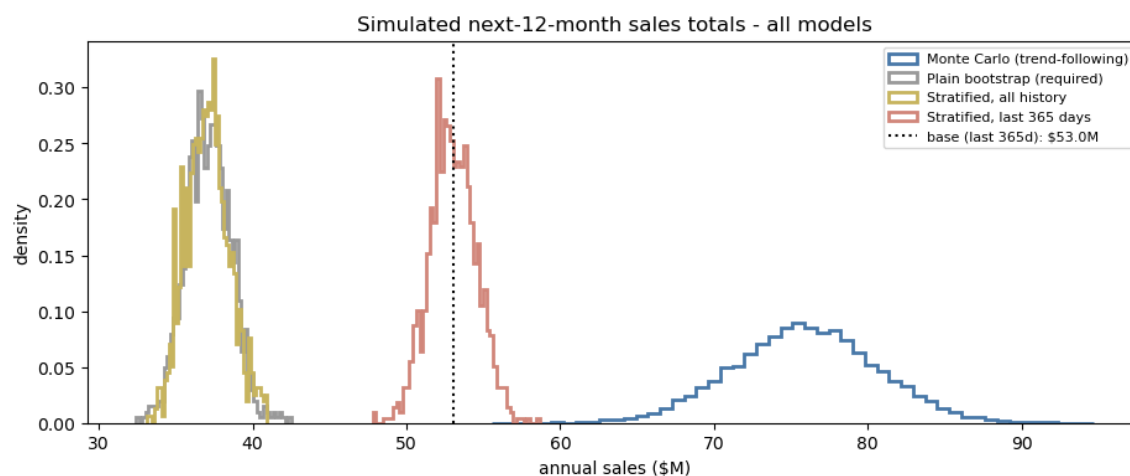


Figure 9 — The four headline distributions. The dotted line marks the current run-rate (last 365 days, \$53.0M).

7.1 Why the models disagree

- **Trend:** the Monte Carlo projects the historical ~43% annual growth forward from the current run-rate; the plain bootstrap implicitly forecasts regression to the three-year average (\$37M). Neither is 'wrong' — they answer different questions ('what if the trend continues?' vs 'what if next year resembles a random mix of the past three?').
- **Risk width:** the bootstrap intervals are 3× narrower because annual totals built from 365 independent daily draws diversify away nearly all variation. The Monte Carlo is more honest about annual-scale uncertainty because the uncertain quantity it models is the annual growth rate itself.
- **Seasonality:** preserved only by the stratified variants; the plain bootstrap's daily profile is flat. For monthly or quarterly phasing, only the stratified models should be used.

7.2 The granularity spectrum — choose the grain by the question

The three resampling grains answer three different questions. Year-grain (Monte Carlo) models the uncertain quantity that matters for targets — annual growth — and gives the most honest annual risk range (\$15.4M wide). Day-grain (stratified/joint-day bootstrap) preserves seasonality and within-day segment mix, so it drives operational phasing and coherent channel/category/territory breakdowns (\$5.0M wide). Transaction-grain (the Poisson simulation) offers the finest drill-down, but every step down in grain adds an independence assumption — days independent of days, order lines independent of order lines — and each added assumption shrinks the simulated annual interval further below the true risk (\$1.5M wide). In short: set targets at year grain, plan operations at day grain, analyse mix at transaction grain.

8. Recommendation

For FY2021 planning, AdventureWorks should work from a forecast range rather than a single number:

- **Main planning range:** \$68M – \$76M — from the Monte Carlo 5th percentile (the budget floor) to its expected value under trend continuation (\$75.9M). Set the official budget near the bottom of this range and the sales target near the top.
- **Downside stress case:** \$53M (range \$50.5M – \$55.5M) — the zero-growth stratified bootstrap; stress-test cash flow and inventory against this scenario.
- **Upside capacity case:** \$84M — the Monte Carlo 95th percentile; do not budget for it, but ensure capacity and supply could serve it.

Quarterly phasing should follow the stratified bootstrap profile in Table 4, and channel/category/territory sub-targets should follow the coherent segment shares in Table 6, scaled to whichever annual scenario management adopts.

8.1 Limitations and next steps

- The growth distribution rests on 12 overlapping rolling fiscal-year observations from a single three-year expansion phase; it cannot see turning points. The Normal heuristic is an assumption, not a finding.
- The bootstrap family assumes days are exchangeable within their stratum; it preserves no week-to-week momentum, and the zero-growth pool freezes the segment mix of the last 12 months.
- Track outcomes (IMPACT): re-run all simulations each quarter as new data arrive — every new quarter adds rolling fiscal-year observations and tightens the growth distribution. Segment-level Monte Carlo (e.g. per channel, with correlated growth draws) is the natural next refinement once more history accumulates.

Appendix — Reproducibility

- Analysis file: AdventureWorks_Simulation_Analysis.ipynb (Python; pandas, numpy, matplotlib, openpyxl).
- All random draws use numpy Generators with fixed seeds (42 for the main sections, an independent stream seeded 7 for the joint-day section) — every number and chart in this report reproduces exactly on re-run.
- Trial counts: 10,000 Monte Carlo trials; 1,000 bootstrap years × 365 days per variant; 1,000 transaction-level simulated years.
- Source data: AdventureWorks Sales.xlsx — Sales_data (fact) joined to Date_data (fiscal calendar), Product_data (category), Sales Territory_data (territory group).

References and Use of AI Tools

- Data: Microsoft AdventureWorks sample dataset (AdventureWorks Sales.xlsx), provided with the course materials.
- Microsoft Learn, “Create a dimensional model report in Power BI Desktop”, <https://learn.microsoft.com/en-us/power-bi/create-reports/desktop-dimensional-model-report> (referenced in the project brief; used to interpret the star-schema structure).
- Richardson, Teeter & Terrell, Data Analytics for Accounting, McGraw-Hill (course textbook; source of the IMPACT framework used to structure this report).
- Software: Python 3 (pandas, NumPy, Matplotlib, openpyxl), Jupyter Notebook, Microsoft Power BI Desktop, Microsoft Word and Excel.
- AI assistance (disclosed per course policy): Claude (Anthropic) was used to assist with implementing the analysis code; the modelling framework and all analytical decisions were specified by the author. The report was written by the author, with AI assistance limited to document formatting and language polishing.